Marcos Morales Maldonado

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EDUCATION

Bachelor of Science in Mechanical Engineering — Anticipated: 06/2029
The University of Texas at El Paso (UTEP) — Overall GPA: 3.91/4.00
Major GPA: 4.00/4.00

Honors & Affiliations:

- Honors Program and Dean's List
- Organizations: The American Society of Mechanical Engineers, SPHE/MAES Community Engagement Committee Member, Secretary of Theta Chi Fraternity.
- Relevant coursework: Graphic & Design Fundamentals, Intro / Electrical / Computer Engineering, Additive Manufacturing Lab.

SKILLS

- Polyglot: English – Professional Working Proficiency, Spanish – Native or Bilingual Proficiency, French – Elementary Proficiency, Japanese – Certificate of N5
- Microsoft Excel (formulas, charts), Word, and PowerPoint
- Understanding of JavaScript
- Skills to lead and work in a team
- Basic knowledge of electrical circuits
- SolidWorks

EXPERIENCE/RELEVANT EXPERIENCE

ORION ROBOTICS — 06/2026–Present, El Paso, TX

- Design and model the rover's chassis and structure in CAD, adapting parts for sensor mounting.
- Support ROS 2 integration of LiDAR and cameras for autonomous navigation and obstacle detection.
- Contribute to building a new robotics team competing in URC and RoboRacer.

RESEARCH S.U.N PROGRAM — 06/2026–Present, El Paso, TX

- Designed 3D drone components with integrated computer vision camera system (MobileNet-SSD).
- Collaborated with team members at S.U.N. (UTEP SHPE/MAES) on drone design and assembly.
- Led 3D design, producing structural printed parts for the drone chassis.

DRAGSTER PROJECT — 03/2026 – 04/2026, El Paso, TX

- Designed a wooden dragster using SolidWorks, applying 3D modeling principles
- Conducted airflow simulations to improve aerodynamic performance
- Built a physical prototype by translating digital designs into hands-on fabrication

SOLAR LAMP PROJECT — 01/2026 – 04/2026, El Paso, TX

- Designed a 3D prototype of a solar-powered lamp based on user needs and market research
- Conducted market analysis and applied design matrices to guide product development
- Developed and presented iterative improvements, working toward a functional prototype